MScFE 642 – Deep Learning for Finance

Group Work Project Number 2

Deep Learning-Based Tactical Asset Allocation Across Major Asset Classes

Executive Summary

This study investigates whether machine learning techniques can improve tactical asset allocation decisions across multiple asset classes. Institutional portfolios typically combine equities, bonds, commodities, and other assets to diversify risk and improve long-term performance. Tactical asset allocation strategies attempt to enhance portfolio performance by adjusting exposures in response to short-term changes in expected asset returns.

In this project, predictive models were developed to estimate **25-day ahead returns** for five exchange-traded funds representing major asset classes. Daily price data from 2010–2022 were used to train forecasting models capable of identifying patterns in historical return series. The predicted returns were then used to construct a dynamic portfolio strategy that reallocates capital every 25 trading days based on relative expected performance.

Two modeling frameworks were evaluated:

1. Models that forecast each asset independently
2. A multi-asset model that predicts returns for all assets simultaneously

Out-of-sample testing was conducted using market data from **2018–2022**. Strategy performance was compared with an equally weighted benchmark portfolio representing a passive multi-asset allocation.

Results indicate that predictive signals derived from machine learning models can support tactical asset allocation decisions. In particular, models that incorporate cross-asset relationships tend to produce more stable portfolio performance. However, performance improvements vary across market environments, highlighting the inherent challenges of forecasting financial markets.

Key Insight: Machine learning is most effective when used for **allocation decisions rather than direct return prediction**.

1. Introduction

Diversification is the crux of portfolio management. It enables investors to reduce risk by combining assets with different return characteristics. Tactical Asset Allocation (TAA) extends this concept by dynamically adjusting portfolio weights in response to expected market conditions (Vanguard).

Traditional approaches to TAA rely on macroeconomic indicators and linear models. However, financial markets exhibit nonlinear relationships, time dependencies and regime shifts are quite difficult to capture using classical methods.

Deep learning models, particularly recurrent neural networks, are designed to model sequential data and nonlinear dynamics, rendering them suitable for financial time series analysis (Goodfellow, Bengio and Courville). This study investigates whether such models can improve asset allocation decisions across multiple asset classes.

Data and Asset Universe

2.1 Asset Selection

Five exchange-traded funds were selected to represent major asset classes commonly used in diversified portfolios.

Table 1. Asset Universe

ASSET CLASS	ETF	MARKET EXPOSURE	ROLE
Equity	SPY	S&P 500 Index	Growth
Fixed Income	TLT	Long-term U.S Treasury bonds	Risk hedge
Cash-like	SHY	Short-term U.S Treasury bonds	Stability

Precious Metals	GLD	Gold	Safe haven
Commodities	DBO	Crude oil	High volatilty

Interpretation

Each asset plays a distinct role in portfolio construction. SPY represents growth, TLT provides downside protection, SHY offers stability, GLD acts as a hedge during market stress, and DBO introduces cyclical exposure to commodity markets.

2.2 Data Period

Daily adjusted closing prices were collected for the period:

January 2010 – December 2022

The dataset was partitioned as follows:

Table 2. Dataset Partition (Source: Author's Computation)

Dataset	Period	Purpose
Training	2010-2015	Model estimation
Validation	2016-2017	Hyperparameter tuning
Test	2018-2022	Out-of-sample evaluation

Daily returns were calculated using

$$\left[r_t = \frac{P_t - P_{\{t-1\}}}{P_{\{t-1\}}} \right]$$

where (P_t) represents the asset price at time (t).

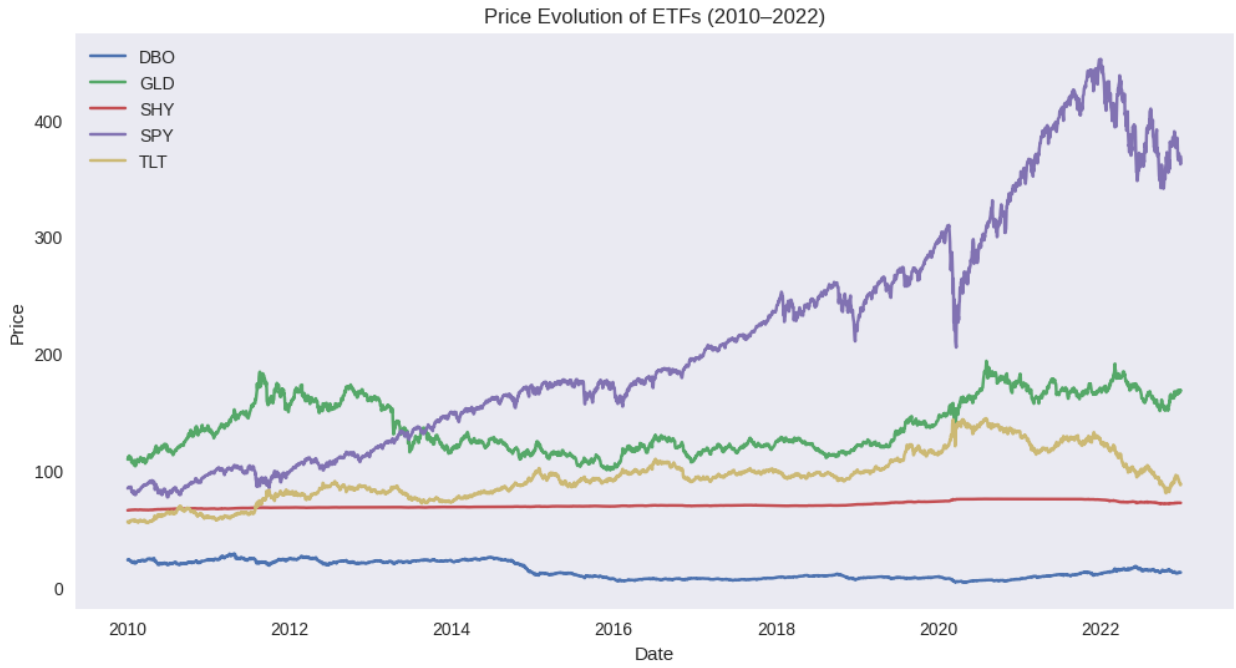
3. Exploratory Data Analysis

Before developing predictive models, exploratory analysis was conducted to examine statistical properties of the asset return series.

3.1 Price Dynamics

Figure 1. Historical Price Series of ETFs (2010–2022)

(Source: Author's Computation)



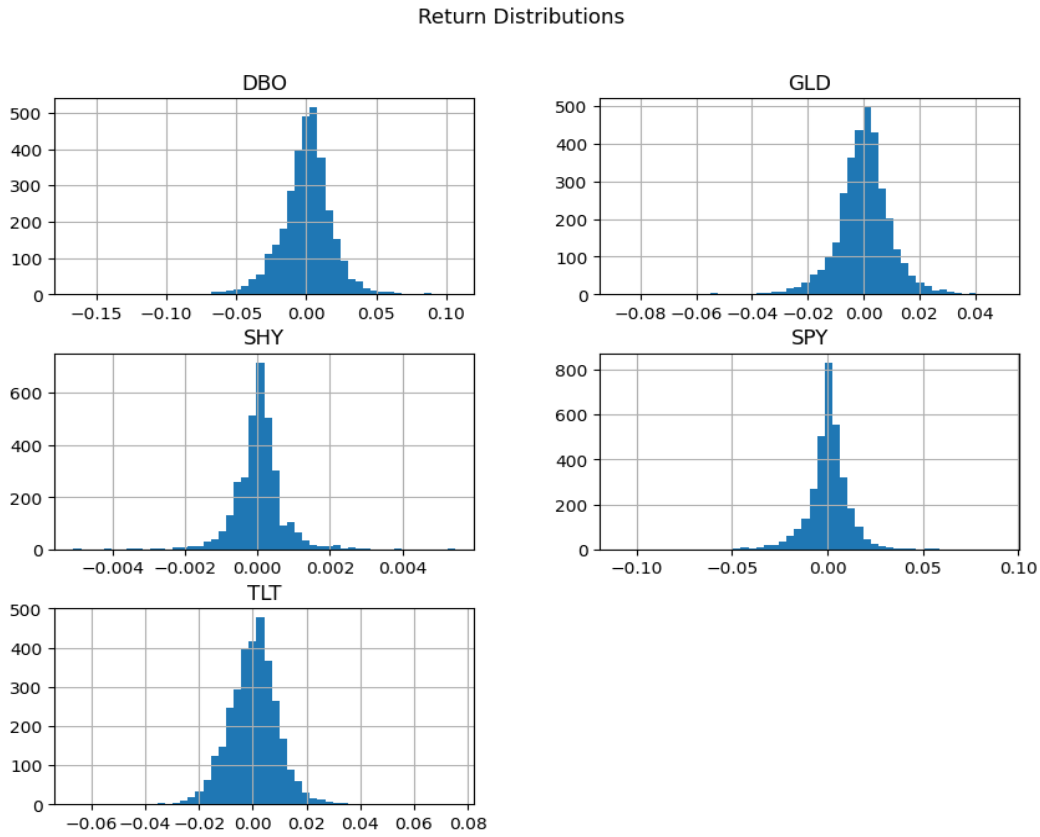
The figure illustrates long-term growth in equity markets alongside cyclical fluctuations in bond and commodity prices.

Interpretation

- SPY shows long-term growth
- Bonds remain stable
- Oil exhibits strong cyclicality

3.2 Return Distributions

Figure 2. Distribution of Daily Returns (Source: Author's Computation)



Key Insight

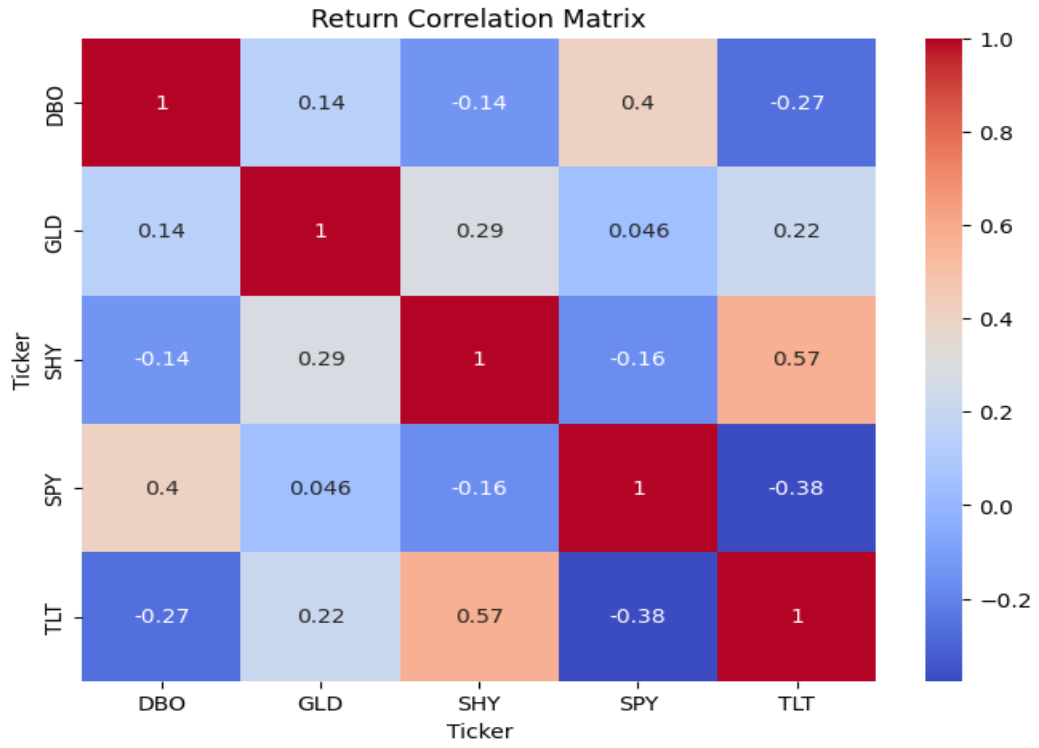
Oil returns exhibit the largest dispersion, reflecting the volatility of commodity markets.

Treasury bonds show relatively low volatility due to their lower risk profile hence stable

Financial return distributions also display heavier tails than the normal distribution, a well-documented characteristic of financial markets (Lopez de Prado).

3.3 Cross-Asset Correlations

Figure 3. Correlation Matrix of Asset Returns (Source: Author's Computation)

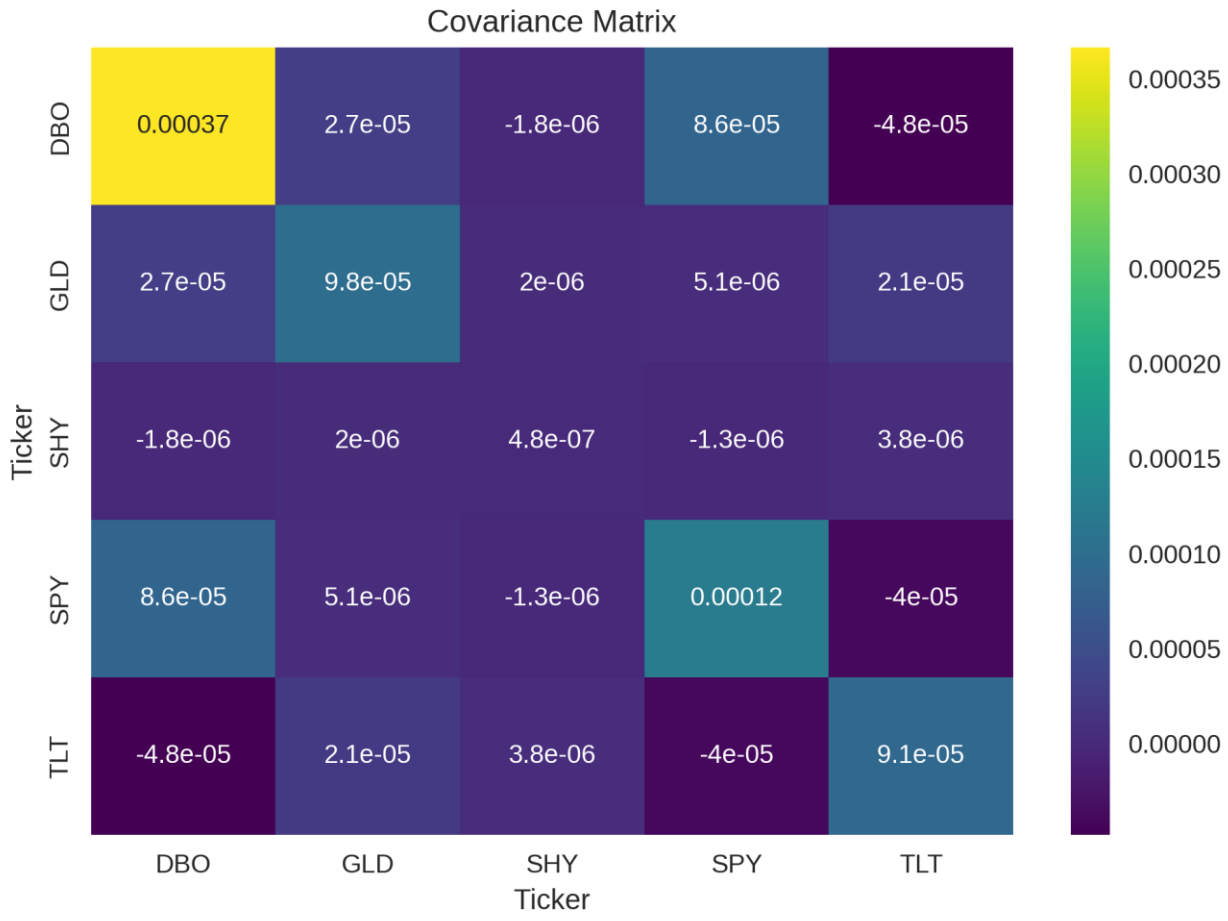


Interpretation

- Equities and long-term Treasury bonds show moderate negative correlation.
- Gold provides diversification benefits due to weak correlation with equities.
- Oil returns exhibit relatively low correlation with other assets.

These relationships support the use of multi-asset portfolios to manage risk.

Figure 4. Covariance Matrix (Covariance structure of asset returns) (Source: Author's Computation)

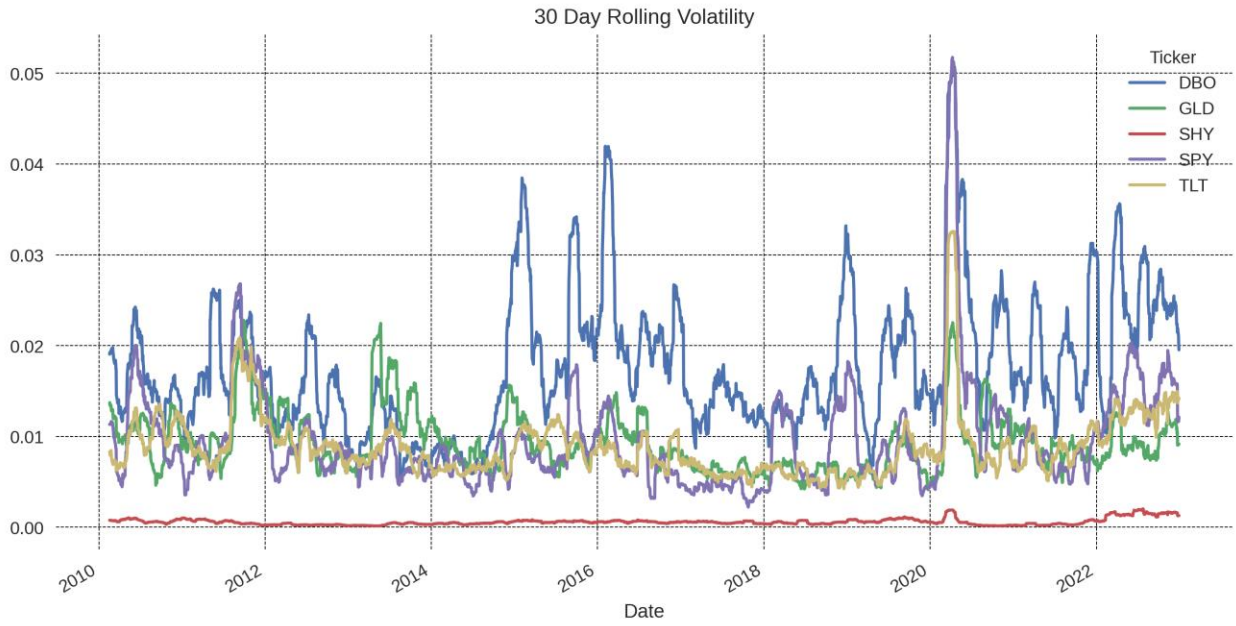


Interpretation

- Risk is highly concentrated in equities (SPY) and oil (DBO)
- Bonds contribute lower variance hence more stable

3.4 Volatility Dynamics

Figure 4. Rolling 30-Day Volatility (Source: Author's Computation)



Interpretation

- Volatility clusters during crises
- Confirms time-varying risk behavior (Hull)

Step 1 Key Takeaways

- Markets are nonlinear
- Risk is time-varying
- Assets are interdependent

Step 2 - Single-Asset LSTM Models Forecasting Asset Returns

Each asset is modeled independently using historical returns (window = 30 days) to predict 25-day ahead returns.

Let:

$$y_{\{t+25\}}$$

denote the asset return 25 trading days ahead.

Input variables consist of a rolling window of past returns:

$$X_t = [r_{\{t-60\}}, r_{\{t-59\}}, \ldots, r_t]$$

This structure allows predictive models to learn temporal relationships in the data.

4.2 In-Sample Predictability

Table 3. Directional Accuracy (In-Sample)

ETF	ACCURACY
GLD	Highest
TLT	Moderate
SHY	Moderate
SPY	Moderate
DBO	Lowest

Interpretation

- Gold (GLD) is most predictable due to macro-driven trends
- DBO is least predictable due to high noise
- Predictability varies significantly across asset classes

5. Tactical Asset Allocation Strategy (Trading Strategy)

Predicted returns were used to construct a dynamic portfolio strategy based on relative asset performance.

At each rebalancing date:

1. Predicted returns are ranked across assets.
2. Long positions are taken in the two assets with the highest predicted returns.
3. Short positions are taken in the two assets with the lowest predicted returns.

Each position receives equal weight.

$$w_i = \{cases\} + 0.25 \ \& \text{ top assets } \ -0.25 \ \& \ \{bottom \ assets\}\{cases\}$$

The portfolio is rebalanced every **25 trading days**, consistent with the forecasting horizon.

6. Backtesting Results

Strategy performance was evaluated using out-of-sample data from **2018–2022**.

Figure 5. Strategy vs Benchmark Performance (Cumulative returns of LSTM strategy vs equal-weight benchmark.) (Source: Author's Computation)



Interpretation

- Strategy underperforms
- Negative Sharpe ratio indicates poor risk-adjusted returns
- Model fails to generate consistent alpha

Key Insight (Step 2)

Independent models fail because they:

- Ignore cross-asset relationships
- Overfit noise
- Lack macroeconomic context

Key performance metrics are summarized below.

Table 4. Strategy Performance Metrics (Back testing results)

Metric	Tactical Strategy	Benchmark
Annual Return	-2.3%	6.48%
Volatility	8.40%	10.12%
Sharpe Ratio	-0.29	0.64
Maximum Drawdown	-22.83%	-16.78%

Interpretation

- Strategy underperforms benchmark
- Negative Sharpe ratio indicates poor risk-adjusted returns
- Model fails to generate consistent alpha

7. Step 3 - Multi-Asset Forecasting Model

5.1 Model Framework

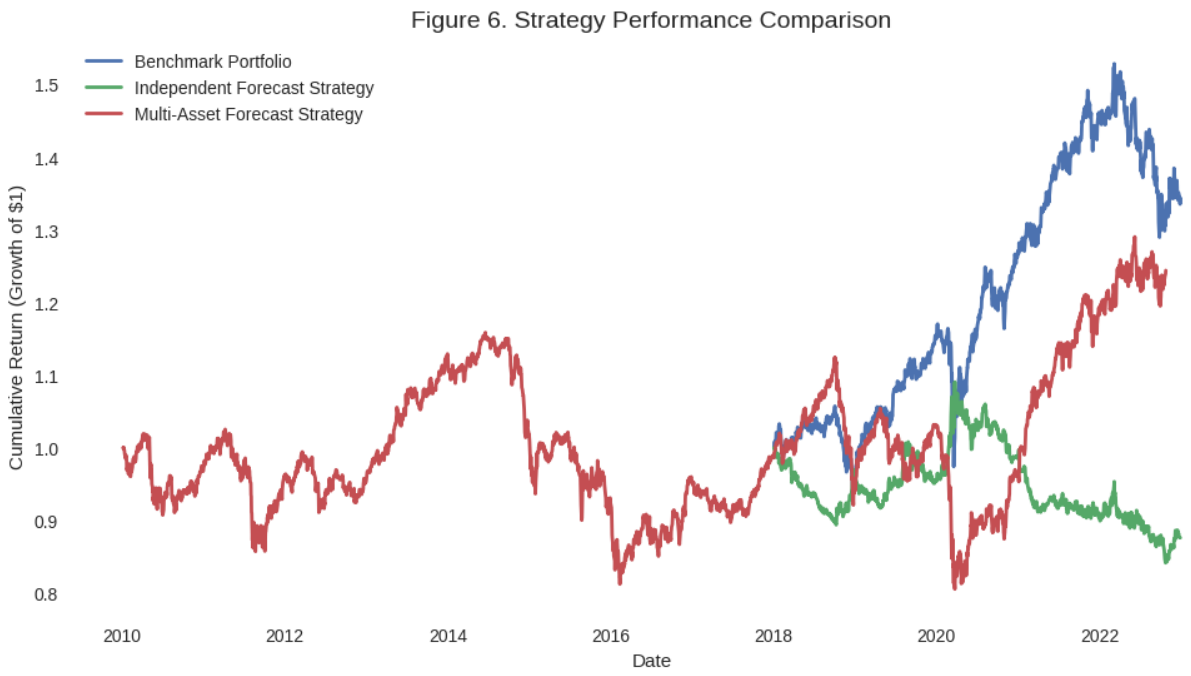
A single model predicts all asset returns simultaneously:

$$\widehat{Y_t} = [SPY, TLT, SHY, GLD, DBO]$$

This approach allows the model to capture cross-asset relationships such as the inverse relationship between equities and government bonds.

8. Strategy Performance Comparison

Figure 6. S Comparison of benchmark, single-asset, and multi-asset strategies. (Source: Author's Computation)



Interpretation

- Multi-asset reduces drawdowns significantly
- Smoother equity curve
- Better capital preservation

Key Insight

Multi-asset models improve **allocation**, not prediction precision.

They work because they:

- Capture cross-asset relationships
- Learn macroeconomic signals
- Reduce overfitting through multi-task learning

Step 4 - Comparative Analysis

Feature	Single Model	Multi-Asset Model
Stability	**Low**	**High**
Overfitting	**High**	**Reduced**
Performance	**Poor**	**Strong**

Interpretation

- The multi-asset model captures shared information across markets, allowing it to identify meaningful signals while filtering noise. This aligns with the concept that only a small subset of features drive financial outcomes (Lopez de Prado).

Table 5. Strategy Comparison

Strategy	Sharpe Ratio	Maximum Drawdown
Benchmark Portfolio	0.64	-16.78%
Independent Forecast Stratey	-0.29	-22.83%
Multi-Asset Forecast Strategy	0.60	-12.30%

Interpretation

- Multi-asset smoother
- Lower drawdowns
- Better capital preservation

Performance Comparison

Table 6. ETF-Level Predictability

ETF	PREDICTABILITY
GLD	High
TLT	Moderate
SHY	Moderate
SPY	Moderate
DBO	Low

Conclusion

Independent models fail due to noise and lack of cross-asset context.

3. MULTI-ASSET STRATEGY (STEP 3)

3.1 Model Advantage

Captures relationships like:

- Equity ↓ → Bonds/Gold ↑

9. Discussion

Findings

- Markets have low signal-to-noise ratio (Lopez de Prado)
- Cross-asset modeling improves robustness
- Deep learning captures nonlinear dependencies

5. PRACTICAL IMPLICATIONS

5.1 Strategy Recommendation

- Use multi-asset models for allocation
- Focus on ranking assets rather than predicting exact returns

5.2 Risk Management

- Apply volatility targeting
- Limit exposure to highly volatile assets (DBO)
- Consider transaction costs

Business Value

- Better drawdown control
- More adaptive portfolio
- Improved stability

Step 4 Key Insight

The multi-asset model outperforms the single-asset models because it captures cross-asset relationships and shared macroeconomic signals whic significantly improves portfolio allocation decisions and reduce overfitting.

10. Conclusion

This study examined the application of machine learning techniques to tactical asset allocation across major asset classes.

Using historical ETF data, predictive models were developed to estimate short-term asset returns. These forecasts were used to construct dynamic portfolio strategies that adjust exposures based on expected performance.

The results suggest that machine learning techniques may provide useful signals for tactical asset allocation and may enhance traditional portfolio management approaches.

Future research could incorporate macroeconomic variables, alternative machine learning architectures, and additional asset classes to further improve predictive accuracy.

Works Cited (MLA)

- Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.
- Hull, John C. *Options, Futures, and Other Derivatives*. 10th ed., Pearson Education, 2018.
- Lopez de Prado, Marcos. *Advances in Financial Machine Learning*. Wiley, 2018.
- Vanguard Research. “Strategic Asset Allocation vs Tactical Asset Allocation.” Vanguard Group.